KAIZEN MATHS

# Lesson starter

**Confidence Intervals**

Margin of Error and Sample Size

Name

**S1**

Find the minimum sample size needed for the required margin of error.

A 90% confidence interval for a population mean must have margin of error at most 1.25. The population standard deviation is 12. Find the minimum sample size.

**S2**

Find the minimum sample size needed for a proportion estimate.

A 99% confidence interval for a proportion should have margin of error at most 0.04. Use planning value p = 0.2. Find the minimum sample size.

**S3**

Find the estimate and margin of error from the interval.

A confidence interval is (93.9, 101.1). Find the point estimate and the margin of error.

**S4**

Compare intervals by thinking about the critical value.

For the same sample data, compare a 95% confidence interval and a 90% confidence interval. Which interval is wider?



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# Exit ticket

**Confidence Intervals**

Margin of Error and Sample Size

Name

**EA1**

Find the minimum sample size needed for the required margin of error.

A 99% confidence interval for a population mean must have margin of error at most 0.5. The population standard deviation is 6. Find the minimum sample size.

**EA2**

Find the minimum sample size needed for a proportion estimate.

A 95% confidence interval for a proportion should have margin of error at most 0.05. Use planning value p = 0.3. Find the minimum sample size.



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Exit ticket

Confidence Intervals

Margin of Error and Sample Size

Name

**EB1**

Find the estimate and margin of error from the interval.

A confidence interval is (89.1, 93.9). Find the point estimate and the margin of error.



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Exit ticket

Confidence Intervals

Margin of Error and Sample Size

Name

EA1

Find the minimum sample size needed for the required margin of error.

A 99% confidence interval for a population mean must have margin of error at most 0.5. The population standard deviation is 6. Find the minimum sample size.

EA2

Find the minimum sample size needed for a proportion estimate.

A 95% confidence interval for a proportion should have margin of error at most 0.05. Use planning value p = 0.3. Find the minimum sample size.

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Confidence Intervals

Margin of Error and Sample Size

Name

EB1

Find the estimate and margin of error from the interval.

A confidence interval is (89.1, 93.9). Find the point estimate and the margin of error.

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# Teacher answers & working

**Confidence Intervals · Margin of Error and Sample Size**

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## S1

Find the minimum sample size needed for the required margin of error.

A 90% confidence interval for a population mean must have margin of error at most 1.25. The population standard deviation is 12. Find the minimum sample size.

**Answer:**

Minimum sample size: n = 250.

**Working:**

For a mean, margin of error $E = z\frac{\sigma}{\sqrt{n}}$.

Rearrange: $n \geq (\frac{z\sigma}{E})^2$.

Substitute z = 1.645, σ = 12, and E = 1.25.

$n \geq (\frac{1.645 \times 12}{1.25})^2 = 249.387$.

Sample size must be a whole number and large enough, so n = 250.

## S2

Find the minimum sample size needed for a proportion estimate.

A 99% confidence interval for a proportion should have margin of error at most 0.04. Use planning value p = 0.2. Find the minimum sample size.

**Answer:**

Minimum sample size: n = 664.

**Working:**

For a proportion, $E = z\sqrt{\frac{p(1-p)}{n}}$.

Rearrange: $n \geq \frac{z^2 p(1-p)}{E^2}$.

Substitute z = 2.576, p = 0.2, and E = 0.04.

$n \geq \frac{2.576^2 \times 0.2 \times 0.8}{0.04^2} = 663.578$.

Round up, so n = 664.

## S3

Find the estimate and margin of error from the interval.

A confidence interval is (93.9, 101.1). Find the point estimate and the margin of error.

**Answer:**

Point estimate = 97.5, margin of error = 3.6.

**Working:**

Point estimate is the midpoint of the interval.

Point estimate = $\frac{93.9+101.1}{2}$ = 97.5.

Margin of error is half the width of the interval.

Margin of error = $\frac{101.1-93.9}{2}$ = 3.6.

# Teacher answers & working

**Confidence Intervals · Margin of Error and Sample Size**

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## S4

Compare intervals by thinking about the critical value.

For the same sample data, compare a 95% confidence interval and a 90% confidence interval. Which interval is wider?

**Answer:**

The 95% confidence interval is wider.

**Working:**

Higher confidence requires a larger critical value.

z for 95% confidence is 1.960.

z for 90% confidence is 1.645.

The larger z value gives a larger margin of error.

Therefore the 95% confidence interval is wider.

## EA1

Find the minimum sample size needed for the required margin of error.

A 99% confidence interval for a population mean must have margin of error at most 0.5. The population standard deviation is 6. Find the minimum sample size.

**Answer:**

Minimum sample size: n = 956.

**Working:**

For a mean, margin of error $E = z\frac{\sigma}{\sqrt{n}}$.

Rearrange: $n \geq (\frac{z\sigma}{E})^2$.

Substitute z = 2.576, σ = 6, and E = 0.5.

$n \geq (\frac{2.576 \times 6}{0.5})^2 = 955.552$.

Sample size must be a whole number and large enough, so n = 956.

## EA2

Find the minimum sample size needed for a proportion estimate.

A 95% confidence interval for a proportion should have margin of error at most 0.05. Use planning value p = 0.3. Find the minimum sample size.

**Answer:**

Minimum sample size: n = 323.

**Working:**

For a proportion, $E = z\sqrt{\frac{p(1-p)}{n}}$.

Rearrange: $n \geq \frac{z^2 p(1-p)}{E^2}$.

Substitute z = 1.960, p = 0.3, and E = 0.05.

$n \geq \frac{1.960^2 \times 0.3 \times 0.7}{0.05^2} = 322.694$.

Round up, so n = 323.

# Teacher answers & working

**Confidence Intervals · Margin of Error and Sample Size**

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**EB1**

Find the estimate and margin of error from the interval.

A confidence interval is (89.1, 93.9). Find the point estimate and the margin of error.

**Answer:**

Point estimate = 91.5, margin of error = 2.4.

**Working:**

Point estimate is the midpoint of the interval.

Point estimate = $\frac{89.1+93.9}{2}$ = 91.5.

Margin of error is half the width of the interval.

Margin of error = $\frac{93.9-89.1}{2}$ = 2.4.